

粗邻近

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摘 要

本文在完全分配格 L 和 M 的框架下研究模糊粗邻近结构. 首先从 (L, M) -模糊囿结构出发定义 (L, M) -模糊粗邻近, 讨论其基本性质以及由粗邻近诱导的模糊邻域系统. 随后引入伪 (L, M) -模糊渐近相似性, 并研究其与 (L, M) -模糊粗结构之间的构造关系.

关键词: (L, M) -模糊囿结构; (L, M) -模糊粗邻近; 模糊渐近相似性

Rough Proximity

Abstract

This paper studies fuzzy coarse proximity structures over completely distributive lattices L and M . It defines (L, M) -fuzzy coarse proximities from (L, M) -fuzzy boundedness structures, investigates the induced fuzzy neighborhood systems, and introduces pseudo (L, M) -fuzzy asymptotic similarities together with their relation to (L, M) -fuzzy coarse structures.

Keywords: (L, M) -fuzzy boundedness structure; (L, M) -fuzzy coarse proximity; fuzzy asymptotic similarity

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第三章 粗邻近的刻画

§3.1 (L, M) -模糊粗邻近

定义3.1.1. 集合 X 上的一个 (L, M) -模糊圈结构是一个映射 $\mathcal{B} : L^X \rightarrow M$, 满足:

$$\text{对 } \forall x \in X, \mathcal{B}\left(\mathcal{X}_{\{x\}}\right) = \top;$$

$$\text{对 } \forall A, B \in L^X, A \leq B, \mathcal{B}(B) \leq \mathcal{B}(A);$$

$$\text{对 } \forall A, B \in L^X, \mathcal{B}(A \vee B) \geq \mathcal{B}(A) \wedge \mathcal{B}(B).$$

则 $(X, \mathcal{B})$ 称为 (L, M) -模糊圈空间, $\mathcal{B}$ 中的元素成为有界的。

注3.1.2. (1) 一个 (L, M) -模糊圈结构满足 $\mathcal{B}(\top_M) = \perp$, 则称为 (L, M) -模糊真圈结构;

(2) 由定义3.1.1可知, 对 $\forall A, B \in L^X, \mathcal{B}(A \vee B) = \mathcal{B}(A) \wedge \mathcal{B}(B)$ 也成立。

例3.1.3. 设 (X, d) 是一个经典的度量空间, 对 $\forall x_0 \in X$, 定义 $\mathcal{B} : [0, 1]^X \rightarrow [0, 1]$,

$$\mathcal{B}(A) = \bigvee_{r>0} \bigwedge_{d(x, x_0)>r} A'(x),$$

则 $(X, \mathcal{B})$ 称为 (L, M) -模糊圈空间。

证明: (1) 对 $\forall y \in X$, 取 $r_0 = d(y, x_0) + 1$, 满足对 $\forall x \in X, d(x, x_0) > r_0$, 则 $x \neq y$, 故 $\mathcal{X}_{\{y\}}(x) = 0$, 则 $\mathcal{X}'_{\{y\}}(x) = 1$, 即 $\bigwedge_{d(x, x_0)>r} \mathcal{X}'_{\{y\}}(x) = 1$,

$$\begin{aligned} \mathcal{B}\left(\mathcal{X}_{\{y\}}\right) &= \bigvee_{r>0} \bigwedge_{d(x, x_0)>r} \mathcal{X}'_{\{y\}}(x) = 1; \end{aligned}$$

(2) 对 $\forall A, B \in L^X, A \leq B$, 则对 $\forall x \in X, A(x) \leq B(x)$, 则 $A'(x) \geq B'(x)$, 所以对给定的 $r > 0$, $\bigwedge_{d(x, x_0)>r} A'(x) \geq \bigwedge_{d(x, x_0)>r} B'(x)$, 故

$$\mathcal{B}(A) = \bigvee_{r>0} \bigwedge_{d(x, x_0)>r} A'(x)$$

$$\geq \bigvee_{r>0} \bigwedge_{d(x,x_0)>r} B'(x) = \mathcal{B}(B);$$

(3) 对 $\forall A, B \in L^X$, 由于

$$\begin{aligned} \mathcal{B}(A) \wedge \mathcal{B}(B) &= \left(\bigvee_{r_1>0} \bigwedge_{d(x,x_0)>r_1} A'(x) \right) \\ &\quad \wedge \left(\bigvee_{r_2>0} \bigwedge_{d(x,x_0)>r_2} B'(x) \right) \\ &= \bigvee_{r_1>0} \bigvee_{r_2>0} \left(\bigwedge_{d(x,x_0)>r_1} A'(x) \wedge \bigwedge_{d(x,x_0)>r_2} B'(x) \right) \end{aligned}$$

取 $r = \max \{ r_1, r_2 \}$, 则

$$\begin{aligned} \bigwedge_{d(x,x_0)>r_1} A'(x) \wedge \bigwedge_{d(x,x_0)>r_2} B'(x) &\leq \bigwedge_{d(x,x_0)>r} A'(x) \wedge \bigwedge_{d(x,x_0)>r} B'(x) \\ &\leq \bigwedge_{d(x,x_0)>r} (A'(x) \wedge B'(x)) \\ &= \bigwedge_{d(x,x_0)>r} (A \vee B)'(x) \end{aligned}$$

故

$$\begin{aligned} \mathcal{B}(A) \wedge \mathcal{B}(B) &= \bigvee_{r_1>0} \bigvee_{r_2>0} \left(\bigwedge_{d(x,x_0)>r_1} A'(x) \wedge \bigwedge_{d(x,x_0)>r_2} B'(x) \right) \\ &\leq \bigvee_{r>0} \bigwedge_{d(x,x_0)>r} (A \vee B)'(x) = \mathcal{B}(A \vee B). \end{aligned}$$

定义3.1.4. $(X, \mathcal{B})$ 是一个 (L, M) -模糊熵空间, 定义映射 $\delta: L^X \times L^X \rightarrow M$ 是 X 上的一个 (L, M) -模糊粗邻近, 若它满足:

$$\text{对 } \forall A, B \in L^X, \delta(A, B) = \delta(B, A);$$

$$\text{对 } \forall A, B \in L^X, \delta(A, B) \leq \mathcal{B}'(A) \wedge \mathcal{B}'(B);$$

$$\text{对 } \forall A, B, C \in L^X, \delta(A \vee B, C) \leq \delta(A, C) \vee \delta(B, C);$$

对 $\forall A, B \in L^X$, $\delta(A, B) \geq \bigwedge_{D \in L^X} (\delta(A, D) \vee \delta(D', B))$;

对 $\forall A, B \in L^X$, $\delta'(A, B) \leq \mathcal{B}(A \wedge B)$.

则 $(X, \mathcal{B}, \delta)$ 称为 (L, M) -模糊粗邻近空间。

注3.1.5. 若 $L = \{0, 1\}$, M 保留为一般的完全分配格, 则退化为石老师定义的 L -值粗邻近; 若 $L, M = \{0, 1\}$, 则退化为经典意义下的粗邻近

注3.1.6. 设 $(X, \mathcal{B}, \delta)$ 为一个 (L, M) -模糊粗邻近空间, 若 $\mathcal{B}(A) = \top$, 则对 $\forall B \in L^X$, $\delta(A, B) = \perp$; 特别地, 若 $\mathcal{B}(\top_M) = \top$, 则 δ 恒为 $\perp$ 。

例3.1.7. 设 $\mathcal{B} : L^X \rightarrow M$ 为非空集合 X 上的一个 (L, M) -模糊真值结构, 定义 $\delta_1, \delta_2 : L^X \times L^X \rightarrow M$ 分别为, 对 $\forall A, B \in L^X$,

$$\begin{aligned}\delta_1(A, B) &= \mathcal{B}'(A \wedge B), \quad \delta_2(A, B) = \mathcal{B}'(A) \\ &\quad \wedge \mathcal{B}'(B),\end{aligned}$$

则称 δ_1 为 (L, M) -模糊离散粗邻近, δ_2 为 (L, M) -模糊平凡粗邻近。

命题3.1.8. 设 $(X, \mathcal{B}, \delta)$ 为一个 (L, M) -模糊粗邻近空间, 对 $\forall A, B, C, D \in L^X$, $A \leq C, B \leq D$, 则 $\delta(A, B) \leq \delta(C, D)$ 。

证明: 由于 $A \leq C, B \leq D$, 则 $A \vee C = C, B \vee D = D$, 故

$$\delta(C, D) = \delta(A \vee C, D) = \delta(A, D) \vee \delta(C, D),$$

所以 $\delta(A, D) \leq \delta(C, D)$, 同理可得 $\delta(A, B) \leq \delta(A, D)$, 故 $\delta(A, B) \leq \delta(C, D)$ 。

命题3.1.9. 设 $(X, \mathcal{B}, \delta)$ 为一个 (L, M) -模糊粗邻近空间, 对 $\forall A, B \in L^X$, 有

$$\delta(A, B) \leq \bigwedge_{D \in L^X} (\delta(A, D) \vee \delta(D', B)).$$

证明: 对 $\forall D \in L^X$, 由于 $B \leq D \vee (B \wedge D')$, 则

$$\begin{aligned}\delta(A, B) &\leq \delta(A, D \vee (B \wedge D')) = \delta(A, D) \\ &\quad \vee \delta(A, B \wedge D'),\end{aligned}$$

而 $\delta'(B, D') \leq \mathcal{B}(B \wedge D')$, 则 $\delta(B, D') \geq \mathcal{B}'(B \wedge D')$, 所以

$$\begin{aligned}\delta(A, B \wedge D') &\leq \mathcal{B}'(A) \wedge \mathcal{B}'(B \wedge D') \\ &\leq \mathcal{B}'(B \wedge D') \leq \delta(B, D'),\end{aligned}$$

故 $\delta(A, B) \leq \delta(A, D) \vee \delta(D', B)$, 得证。

定义3.1.10. 设 $(X, \mathcal{B}, \delta)$ 为一个 (L, M) -模糊粗邻近空间, 定义映射 $N_\delta : L^X \times L^X \rightarrow M$, 对 $\forall A, B \in L^X$, $N_\delta(A, B) = (\delta(A, B'))'$, N_δ 称为由 δ 诱导的 (L, M) -模糊粗邻域。

定理3.1.11. 设 $(X, \mathcal{B}, \delta)$ 为一个 (L, M) -模糊粗邻近空间, N_δ 称为由 δ 诱导的 (L, M) -模糊粗邻域, 则: 对 $\forall A, B, C \in L^X$

- (1) $\mathcal{B}(A) \leq N_\delta(A, B)$;
- (2) $N_\delta(A, B) \leq \mathcal{B}(A \wedge B')$;
- (3) 若 $A \leq B$, 则 $N_\delta(B, C) \leq N_\delta(A, C)$;
- (4) 若 $B \leq C$, 则 $N_\delta(A, B) \leq N_\delta(A, C)$;
- (5) $\mathcal{B}(A \wedge B') \wedge N_\delta(B, C) \leq N_\delta(A, C)$;
- (6) $N_\delta(A, B) \wedge N_\delta(B, C) \leq N_\delta(A, C)$;
- (7) $N_\delta(A, B) \wedge N_\delta(A, B') \leq \mathcal{B}(A)$;
- (8) $N_\delta(A, B) = N_\delta(B', A')$ 。

证明: (1) 对 $\forall A, B' \in L^X$, 由于 $\delta(A, B') \leq \mathcal{B}'(A) \wedge \mathcal{B}'(B')$, 所以 $\delta(A, B') \leq \mathcal{B}'(A)$, 故

$$N_\delta(A, B) = (\delta(A, B'))' \geq \mathcal{B}(A);$$

(2) 对 $\forall A, B \in L^X$, 由于 $\delta'(A, B) \leq \mathcal{B}(A \wedge B)$, 则 $N_\delta(A, B) = \delta'(A, B') \leq \mathcal{B}(A \wedge B')$;

(3) 对 $\forall A, B \in L^X$, $A \leq B$, 则 $\delta(A, C') \leq \delta(B, C')$, 故

$$N_\delta(A, C) = \delta'(A, C') \geq \delta'(B, C') = N_\delta(B, C);$$

(4) 与 (3) 同理;

(5) 对 $\forall A, B \in L^X$, 由于 $A \leq B \vee (A \wedge B')$, 则

$$\begin{aligned} \delta(A, C') &\leq \delta(B \vee (A \wedge B'), C') = \delta(B, C') \\ &\vee \delta(A \wedge B', C') \end{aligned}$$

又

$$\begin{aligned} \delta(A \wedge B', C') &\leq \mathcal{B}'(A \wedge B') \wedge \mathcal{B}'(C') \\ &\leq \mathcal{B}'(A \wedge B'), \end{aligned}$$

则 $\delta(A, C') \leq \delta(B, C') \vee \mathcal{B}'(A \wedge B')$, 即

$$\begin{aligned} N_\delta(A, C) &= \delta'(A, C') \geq \delta'(B, C') \wedge \mathcal{B}(A \wedge B') \\ &= N_\delta(B, C) \wedge \mathcal{B}(A \wedge B'); \end{aligned}$$

(6) 对 $\forall A, B, C \in L^X$, 由于 $\delta(A, C') = \bigwedge_{E \in L^X} (\delta(A, E) \vee \delta(E', C'))$, 取 $E = B'$, 则 $\delta(A, C') \leq \delta(A, B') \vee \delta(B, C')$, 故

$$\begin{aligned} N_\delta(A, C) &= \delta'(A, C') \geq \delta'(A, B') \wedge \delta'(B, C') \\ &= N_\delta(A, B) \wedge N_\delta(B, C); \end{aligned}$$

(7) 由于 $N_\delta(A, B) \wedge N_\delta(B, A') \leq N_\delta(A, A')$, 而

$$N_\delta(B, A') = \delta'(B, A) = \delta'(A, B), \text{ 同理}$$

$$N_\delta(A, B') = \delta'(A, B), \text{ 则}$$

$$N_\delta(A, B') = N_\delta(B, A'), \text{ 故}$$

$$N_\delta(A, B) \wedge N_\delta(A, B') \leq N_\delta(A, A'), \text{ 又由于}$$

$N_\delta(A, A') = \delta'(A, A) \leq \mathcal{B}(A \wedge A) = \mathcal{B}(A)$, 则

$$N_\delta(A, B) \wedge N_\delta(A, B') \leq \mathcal{B}(A);$$

(8) 对 $\forall A, B \in L^X$, $N_\delta(B', A') = \delta'(B', A) = \delta'(A, B') = N_\delta(A, B)$ 。

注3.1.12. 设 $(X, \mathcal{B}, \delta)$ 为一个 (L, M) -模糊粗邻近空间, $N_\delta : L^X \times L^X \rightarrow M$ 是由 δ 诱导的 (L, M) -模糊粗邻域, 则对 $\forall A, B, D \in L^X$, $\mathcal{B}(D) \wedge N_\delta(A, B) \leq N_\delta(A \vee D, B)$ 。

定理3.1.13. 映射 $\mathcal{B} : L^X \rightarrow M$, 是非空集合 X 上的一个 (L, M) -模糊真圈结构, 映射 $\delta : L^X \times L^X \rightarrow M$ 满足定义3.1.4里的 (1) - (3), (5), 设 $N_\delta : L^X \times L^X \rightarrow M$ 是由 $N_\delta(A, B) = (\delta(A, B'))'$ 定义的 (L, M) -模糊粗邻域, 则以下两条等价:

(1) 对 $\forall A, B \in L^X$, $\delta(A, B) \geq \bigwedge_{D \in L^X} (\delta(A, D) \vee \delta(D', B))$; (定义3.1.4的 (4))

(2) 对 $\forall A, B \in L^X$, $N_\delta(A, B) \leq \bigvee_{C \in L^X} (N_\delta(A, C) \wedge N_\delta(C, B))$;

证明: (1) 推 (2): 对 $\forall A, B \in L^X$, 由于 $\delta(A, B') \geq \bigwedge_{D \in L^X} (\delta(A, D) \vee \delta(D', B'))$, 则 $\delta'(A, B') \leq \bigvee_{D \in L^X} (\delta'(A, D) \vee \delta'(D', B'))$, 所以

$$\begin{aligned} N_\delta(A, B) &\leq \bigvee_{D \in L^X} (N_\delta(A, D') \wedge N_\delta(D', B)) \\ &\leq \bigvee_{C \in L^X} (N_\delta(A, C) \wedge N_\delta(C, B)); \end{aligned}$$

(2) 推 (1): 对 $\forall A, B \in L^X$, 由于 $N_\delta(A, B') \leq \bigvee_{C \in L^X} (N_\delta(A, C) \wedge N_\delta(C, B'))$, 则 $\delta'(A, B) \leq \bigvee_{C \in L^X} (\delta'(A, C') \vee \delta'(C, B))$, 所以

$$\begin{aligned} \delta(A, B) &\geq \bigwedge_{C \in L^X} (\delta(A, C') \vee \delta(C, B)) \\ &\geq \bigwedge_{D \in L^X} (\delta(A, D) \vee \delta(D', B)). \end{aligned}$$

§3.2 伪 (L, M) -模糊渐近相似性

定义3.2.1. 设 X 为一个非空集合。映射 $\varnothing : L^X \times L^X \rightarrow M$ 称为 L^X 上的一个 (L, M) -模糊等价关系, 若它满足以下条件: 对 $\forall A, B, C \in L^X$,

(1) 自反性: $\varnothing(A, A) = \top$;

(2) 对称性: $\varnothing(A, B) = \varnothing(B, A)$;

(3) 传递性: $\varnothing(A, B) \wedge \varnothing(B, C) \leq \varnothing(A, C)$ 。

我们称 L^X 上的一个 (L, M) -模糊等价关系 $\varnothing$ 为 X 上的一个伪 (L, M) -模糊渐近相似性, 若它满足以下条件:

(4) 对 $\forall A, B, C, D \in L^X$,

$$\varnothing(A, B) \wedge \varnothing(C, D) \leq \varnothing(A \vee C, B \vee D)。$$

进一步地, 称伪 (L, M) -模糊渐近相似性 $\varnothing$ 为 X 上的一个 (L, M) -模糊渐近相似性, 若它满足以下分解条件:

(5) 对 $\forall A_1, A_2, B_1, B_2 \in L^X$,

$$\begin{aligned} \varnothing(A_1 \vee A_2, B) &\leq \bigvee_{B_1 \vee B_2 = B} \varnothing(A_1, B_1) \\ &\wedge \varnothing(A_2, B_2)。 \end{aligned}$$

例3.2.2. 设 X 为一个非空集合, 定义映射 $\varnothing_D, \varnothing_I : L^X \times L^X \rightarrow M$ 为, 对 $\forall A, B \in L^X$:

$$\begin{aligned} \varnothing_D(A, B) &= \begin{cases} \perp, & A \neq B \\ \top, & A = B \end{cases} ; \varnothing_I(A, B) \\ &= \top, \end{aligned}$$

则 $\varnothing_D, \varnothing_I$ 都是 L^X 上的一个 (L, M) -模糊渐近相似性, 它们分别称为 L^X 上的离散/平凡 (L, M) -模糊渐近相似性。

定理3.2.3. 设 $(X, \mathcal{B}, \delta)$ 为一个 (L, M) -模糊粗邻近空间, 定义映射 $\varnothing_\delta : L^X \times L^X \rightarrow M$ 满足:

$$\begin{aligned} \varnothing_\delta(A, B) &= \left(\bigwedge_{A_1 \leq A} \mathcal{B}(A_1) \vee \delta(A_1, B) \right) \\ &\wedge \left(\bigwedge_{B_1 \leq B} \mathcal{B}(B_1) \vee \delta(A, B_1) \right), \end{aligned}$$

则 $\varnothing_\delta$ 称为 X 上的一个伪 (L, M) -模糊渐近相似性。

证明：先证 $\varnothing_\delta$ 为 L^X 上的一个 (L, M) -模糊等价关系：对 $\forall A, B, C, D \in L^X$,

(1) 由于 $\delta(A, B) = \delta(B, A)$, 所以显然 $\varnothing_\delta(A, B) = \varnothing_\delta(B, A)$;

(2) 对 $\forall A_1 \leq A$, 由于 $\delta'(A_1, A) \leq \mathcal{B}(A_1 \wedge A) = \mathcal{B}(A_1)$, 则 $\delta(A_1, A) \geq \mathcal{B}'(A_1)$,

故

$$\mathcal{B}(A_1) \vee \delta(A_1, A) \geq \mathcal{B}(A_1)$$

$$\vee \mathcal{B}'(A_1) = \top,$$

同理, $\mathcal{B}(A) \vee \delta(A, A_1) = \top$, 则 $\varnothing_\delta(A, A) = \top$;

(3) 对 $\forall A_1 \leq A$, 由于 $B \leq D \vee (B \wedge D')$, 则

$$\delta(A_1, B) \leq \delta(A_1, D \vee (B \wedge D')) = \delta(A_1, D)$$

$$\vee \delta(A_1, B \wedge D'),$$

又因为

$$\delta(A_1, B \wedge D') \leq \mathcal{B}'(A_1) \wedge \mathcal{B}'(B \wedge D')$$

$$\leq \mathcal{B}'(B \wedge D'),$$

所以 $\delta(A_1, B) \leq \delta(A_1, D) \vee \mathcal{B}'(B \wedge D')$, 则

$$\begin{aligned} \varnothing_\delta(A, B) &= \left(\bigwedge_{A_1 \leq A} \mathcal{B}(A_1) \vee \delta(A_1, B) \right) \\ &\wedge \left(\bigwedge_{B_1 \leq B} \mathcal{B}(B_1) \vee \delta(A, B_1) \right) \\ &\leq \mathcal{B}(A_1) \vee \delta(A_1, B) \leq \mathcal{B}(A_1) \\ &\vee \delta(A_1, D) \vee \mathcal{B}'(B \wedge D'), \end{aligned}$$

由于 $B \wedge D' \leq B$, 则

$$\varnothing_\delta(B, C) = \left(\bigwedge_{B_1 \leq B} \mathcal{B}(B_1) \vee \delta(B_1, C) \right)$$

$$\begin{aligned} \wedge \left(\bigwedge_{C_1 \leq C} \mathcal{B}(C_1) \vee \delta(B, C_1) \right) \\ \leq \mathcal{B}(B \wedge D') \vee \delta(B \wedge D', C), \end{aligned}$$

而 $B \wedge D' \leq D'$, 则 $\delta(B \wedge D', C) \leq \delta(D', C)$, 故 $\varnothing_\delta(B, C) \leq \mathcal{B}(B \wedge D') \vee \delta(D', C)$,
将上面得到的两个式子相交可得:

$$\begin{aligned} \varnothing_\delta(A, B) \wedge \varnothing_\delta(B, C) \\ \leq (\mathcal{B}(A_1) \vee \delta(A_1, D) \vee \mathcal{B}'(B \wedge D')) \\ \wedge (\mathcal{B}(B \wedge D') \vee \delta(D', C)), \end{aligned}$$

故

$$\begin{aligned} \varnothing_\delta(A, B) \wedge \varnothing_\delta(B, C) &\leq \mathcal{B}(A_1) \\ &\vee \delta(A_1, D) \vee \delta(D', C), \end{aligned}$$

则

$$\begin{aligned} \varnothing_\delta(A, B) \wedge \varnothing_\delta(B, C) &\leq \mathcal{B}(A_1) \\ \vee \bigvee_{D \in L^X} (\delta(A_1, D) \vee \delta(D', C)) \\ &= \mathcal{B}(A_1) \vee \delta(A_1, C), \end{aligned}$$

同理可得, 都 $\forall C_1 \leq C$,

$$\begin{aligned} \varnothing_\delta(A, B) \wedge \varnothing_\delta(B, C) &\leq \mathcal{B}(C_1) \\ &\vee \delta(A, C_1), \end{aligned}$$

则

$$\begin{aligned} \varnothing_\delta(A, B) \wedge \varnothing_\delta(B, C) \\ \leq \left(\bigwedge_{A_1 \leq A} \mathcal{B}(A_1) \vee \delta(A_1, C) \right) \end{aligned}$$

$$\begin{aligned} \wedge \left(\bigwedge_{C_1 \leq C} \mathcal{B}(C_1) \vee \delta(A, C_1) \right) \\ = \varnothing_\delta(A, C), \end{aligned}$$

得证 $\varnothing_\delta$ 为 L^X 上的一个 (L, M) -模糊等价关系；

下证对 $\forall A, B, C, D \in L^X$,

$$\begin{aligned} \varnothing_\delta(A, B) \wedge \varnothing_\delta(C, D) \\ \leq \varnothing_\delta(A \vee C, B \vee D): \end{aligned}$$

令 $A \vee C = U$, $B \vee D = V$, 对 $\forall U_1 \leq U$, 由于 $U_1 = (U_1 \wedge A) \vee (U_1 \wedge C)$, 记 $U_A = U_1 \wedge A \leq A$, $U_C = U_1 \wedge C \leq C$, 则

$$\begin{aligned} \varnothing_\delta(A, B) &= \left(\bigwedge_{A_1 \leq A} \mathcal{B}(A_1) \vee \delta(A_1, B) \right) \\ \wedge \left(\bigwedge_{B_1 \leq B} \mathcal{B}(B_1) \vee \delta(A, B_1) \right) \\ &\leq \mathcal{B}(U_A) \vee \delta(U_A, B), \\ \varnothing_\delta(C, D) &= \left(\bigwedge_{C_1 \leq C} \mathcal{B}(C_1) \vee \delta(C_1, D) \right) \\ \wedge \left(\bigwedge_{D_1 \leq D} \mathcal{B}(D_1) \vee \delta(C, D_1) \right) \\ &\leq \mathcal{B}(U_C) \vee \delta(U_C, D), \end{aligned}$$

又由于 $B \leq V$, $D \leq V$, 则 $\delta(U_A, B) \leq \delta(U_A, V)$, $\delta(U_C, D) \leq \delta(U_C, V)$, 两式相交可得

$$\begin{aligned} \varnothing_\delta(A, B) \wedge \varnothing_\delta(C, D) \\ \leq (\mathcal{B}(U_A) \vee \delta(U_A, V)) \\ \wedge (\mathcal{B}(U_C) \vee \delta(U_C, V)) \\ = (\mathcal{B}(U_A) \wedge \mathcal{B}(U_C)) \end{aligned}$$

$$\begin{aligned}
& \vee (\mathcal{B}(U_A) \wedge \delta(U_C, V)) \\
& \vee (\delta(U_A, V) \wedge \mathcal{B}(U_C)) \\
& \vee (\delta(U_A, V) \wedge \delta(U_C, V)) \\
& \leq (\mathcal{B}(U_A) \wedge \mathcal{B}(U_C)) \vee \delta(U_C, V) \\
& \vee \delta(U_A, V)
\end{aligned}$$

又因为

$$\begin{aligned}
\mathcal{B}(U_A) \wedge \mathcal{B}(U_C) &= \mathcal{B}(U_A \vee U_C) \\
&= \mathcal{B}(U_1), \delta(U_C, V) \vee \delta(U_A, V) \\
&= \delta(U_C \vee U_A, V) = \delta(U_1, V),
\end{aligned}$$

则

$$\begin{aligned}
\varnothing_\delta(A, B) \wedge \varnothing_\delta(C, D) &\leq \mathcal{B}(U_1) \\
&\vee \delta(U_1, V),
\end{aligned}$$

同理对 $\forall V_1 \leq V$,

$$\begin{aligned}
\varnothing_\delta(A, B) \wedge \varnothing_\delta(C, D) &\leq \mathcal{B}(V_1) \\
&\vee \delta(U, V_1),
\end{aligned}$$

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$$\begin{aligned}
\varnothing_\delta(A, B) \wedge \varnothing_\delta(C, D) &\leq \varnothing_\delta(U, V) \\
&= \varnothing_\delta(A \vee C, B \vee D).
\end{aligned}$$

定理3.2.4. L^X 上的一个 (L, M) -模糊等价关系 $\varnothing$ 为 X 上的一个伪 (L, M) -模糊渐近相似性, 定义 $\mathcal{C}_\varnothing : L^{X \times X} \rightarrow M$ 满足, 对 $\forall E \in L^{X \times X}$, $\mathcal{C}_\varnothing(E) = \bigwedge_{F \leq E} \varnothing(D_F, R_F)$, 其中 $D_F(x) = \bigvee_{y \in X} F(x, y)$, $R_F(y) = \bigvee_{x \in X} F(x, y)$, 则称 $\mathcal{C}_\varnothing$ 为 L^X 上的一个 (L, M) -模糊粗结构。

证明： (1) 由于 $\Delta_X = \Delta_X^{-1}$, 则

$$\begin{aligned}\mathcal{C}_\emptyset(\Delta_X) &= \bigwedge_{F \leq \Delta_X^{-1}} \emptyset(D_F, R_F) \\ &= \bigwedge_{F \leq \Delta_X} \emptyset(D_F, R_F),\end{aligned}$$

对 $\forall F \leq \Delta_X$, 则 $F(x, y) \leq \Delta_X(x, y)$, 当 $x \neq y$ 时, $\Delta_X(x, y) = \perp$, 则 $F(x, y) = \perp$, $\mathcal{C}_\emptyset(\Delta_X) = \emptyset(\perp, \perp) = \top$; 当 $x = y$ 时, $D_F(x) = \bigvee_{y \in X} F(x, y) = F(x, x)$, $R_F(y) = \bigvee_{x \in X} F(x, y) = F(x, x)$, 则 $\emptyset(D_F, R_F) = \top$, 故 $\mathcal{C}_\emptyset(\Delta_X) = \top$;

(2) 对 $\forall E, G \in L^{X \times X}$, $E \leq G$, 即 $\forall F \leq E$, 一定有 $F \leq G$, 显然 $\mathcal{C}_\emptyset(E) \geq \mathcal{C}_\emptyset(G)$;

(3) 由于 $\mathcal{C}_\emptyset(E \vee G) = \bigwedge_{F \leq E \vee G} \emptyset(D_F, R_F)$, 对 $\forall F \leq E \vee G$, 令 $F_E = F \wedge E$, $F_G = F \wedge G$, 则 $F_E \leq E$, $F_G \leq G$, 且 $F = F \wedge (E \vee G) = F_E \vee F_G$, 则 $D_F = D_{F_E \vee F_G} = D_{F_E} \vee D_{F_G}$, $R_F = R_{F_E \vee F_G} = R_{F_E} \vee R_{F_G}$, 故

$$\begin{aligned}\emptyset(D_{F_E}, R_{F_E}) \wedge \emptyset(D_{F_G}, R_{F_G}) \\ \leq \emptyset(D_{F_E} \vee D_{F_G}, R_{F_E} \vee R_{F_G}) = \emptyset(D_F, R_F),\end{aligned}$$

由于 $F_E \leq E$, $F_G \leq G$, 则 $\mathcal{C}_\emptyset(E) \leq \emptyset(D_{F_E}, R_{F_E})$, $\mathcal{C}_\emptyset(G) \leq \emptyset(D_{F_G}, R_{F_G})$, 故

$$\begin{aligned}\mathcal{C}_\emptyset(E) \wedge \mathcal{C}_\emptyset(G) \\ \leq \emptyset(D_{F_E}, R_{F_E}) \wedge \emptyset(D_{F_G}, R_{F_G}) \\ \leq \emptyset(D_F, R_F),\end{aligned}$$

$$\begin{aligned}\mathcal{C}_\emptyset(E) \wedge \mathcal{C}_\emptyset(G) \\ \leq \bigwedge_{F \leq E \vee G} \emptyset(D_F, R_F) = \mathcal{C}_\emptyset(E \vee G);\end{aligned}$$

(4) 对 $\forall F \leq E \circ G$, 即 $F(x, y) \leq \bigvee_{z \in X} (E(x, z) \wedge G(z, y))$, 令

$$F_E(x, z) = \bigvee_{y \in X} (F(x, y) \wedge E(x, z) \wedge G(z, y)), \quad F_G(z, y)$$

$$= \bigvee_{x \in X} (F(x, y) \wedge E(x, z) \wedge G(z, y)),$$

显然 $F_E \leq E$, $F_G \leq G$, 又因为

$$\begin{aligned} D_{F_E}(x) &= \bigvee_{z \in X} F_E(x, z) \\ &= \bigvee_{z, y \in X} (F(x, y) \wedge E(x, z) \wedge G(z, y)) \\ &= \bigvee_{y \in X} F(x, y) = D_F(x), \end{aligned}$$

所以 $D_{F_E} = D_F$, 同理可得 $R_{F_G} = R_F$, 又因为

$$\begin{aligned} R_{F_E}(z) &= \bigvee_{x \in X} F_E(x, z) \\ &= \bigvee_{x, y \in X} (F(x, y) \wedge E(x, z) \wedge G(z, y)), \end{aligned}$$

$$\begin{aligned} D_{F_G}(z) &= \bigvee_{y \in X} F_G(z, y) \\ &= \bigvee_{x, y \in X} (F(x, y) \wedge E(x, z) \wedge G(z, y)), \end{aligned}$$

则 $R_{F_E} = D_{F_G}$, 由于 $F_E \leq E$ 及 $F_G \leq G$, 则

$$\begin{aligned} \mathcal{C}_\emptyset(E) &\leq \emptyset(D_{F_E}, R_{F_E}) \\ &= \emptyset(D_F, R_{F_E}), \mathcal{C}_\emptyset(G) \\ &\leq \emptyset(D_{F_G}, R_{F_G}) = \emptyset(D_{F_G}, R_F), \end{aligned}$$

故

$$\begin{aligned} \mathcal{C}_\emptyset(E) \wedge \mathcal{C}_\emptyset(G) &\leq \emptyset(D_F, R_{F_E}) \wedge \emptyset(D_{F_G}, R_F) \\ &\leq \emptyset(D_F, R_F), \end{aligned}$$

则

$$\mathcal{C}_\emptyset(E) \wedge \mathcal{C}_\emptyset(G)$$

$$\leq \bigwedge_{F \leq E \circ G} \varnothing(D_F, R_F) = \mathcal{C}_\varnothing(E \circ G);$$

(5) 对 $F \leq E^{-1}$, 有 $F(x, y) \leq E^{-1}(x, y)$, 则 $F^{-1}(y, x) \leq E^{-1}(x, y) = E(y, x)$, 即 $F^{-1} \leq E$, 而

$$\begin{aligned} R_{F^{-1}}(y) &= \bigvee_{x \in X} F^{-1}(x, y) = \bigvee_{x \in X} F(y, x) \\ &= D_F(y), \end{aligned}$$

则

$$\begin{aligned} \varnothing(D_F, R_F) &= \varnothing(R_{F^{-1}}, D_{F^{-1}}) \\ &= \varnothing(D_{F^{-1}}, R_{F^{-1}}), \end{aligned}$$

故

$$\begin{aligned} \mathcal{C}_\varnothing(E^{-1}) &= \bigwedge_{F \leq E^{-1}} \varnothing(D_F, R_F) \\ &= \bigwedge_{F^{-1} \leq E} \varnothing(D_{F^{-1}}, R_{F^{-1}}) = \mathcal{C}_\varnothing(E). \end{aligned}$$